video

S_AXI_LITE
s00_axi

TMDS

ctrl

M00_AXI

s_axi_aclk

aclk

s00_axi_aresetn

mm2s_introut hdmi_oen
irq

aresetn

FIXED_IO

GPIO_0[31:0]

IIC_1

LCD_GPIO

LCD_SPI

M_AXI_0 M_AXI_0_ARESETN[0:0]

Vaux1 Vaux5 Vaux6 Vaux9 Vaux13 Vaux15 Vp_Vn

arduino_direct_iic arduino_direct_spi arduino_gpio

axi_protocol_convert_0

AXI Protocol Converter

S_AXI aclk M_AXI aresetn

axi_quad_spi_0

AXI Quad SPI

SPI_0 AXI_LITE

ext_spi_clk
s_axi_aclk
s_axi_aresetn

ip2intc_irpt

btns_gpio

AXI GPIO

S_AXI GPIO
s_axi_aclk
s_axi_aresetn

ip2intc_irpt

btns_4bits

codec_addr[1:0] hdmi_in_ddc hdmi_in_hpd[0:0]

hdmi_out

hdmi_out_hpd[0:0]

lcd_gpio

AXI GPIO

S_AXI s_axi_aclk GPIO
s_axi_aresetn

leds_gpio

AXI GPIO

S_AXI GPIO
s_axi_aclk
s_axi_aresetn

ip2intc_irpt

leds_4bits

pmoda_rpi_gpio
pmodb_gpio

M_AXI_0_CLK

audio_clk_10MHz bclk
lrclk

sdata_i

sdata_o

ps7_0

ZYNQ7 Processing System

GPIO_0 DDR
FIXED_IO

IIC_1 USBIND_0 S_AXI_HP0_FIFO_CTRL M_AXI_GP0 M_AXI_GP1

S_AXI_HP0 M_AXI_GP0_ACLK M_AXI_GP1_ACLK

S_AXI_HP0_ACLK

IRQ_F2P[2:0] FCLK_CLK0 FCLK_CLK1 FCLK_CLK2 FCLK_CLK3 FCLK_RESET0_N

ps7_0_axi_periph

AXI Interconnect

S00_AXI

M00_AXI M01_AXI M02_AXI M03_AXI M04_AXI M05_AXI M06_AXI M07_AXI

ACLK
ARESETN
S00_ACLK

S00_ARESETN M00_ACLK M00_ARESETN M01_ACLK M01_ARESETN M02_ACLK M02_ARESETN M03_ACLK M03_ARESETN M04_ACLK M04_ARESETN M05_ACLK M05_ARESETN M06_ACLK M06_ARESETN M07_ACLK M07_ARESETN

ps7_0_axi_periph1

AXI Interconnect

S00_AXI

M00_AXI M01_AXI M02_AXI M03_AXI M04_AXI

ACLK
ARESETN

S00_ACLK

S00_ARESETN M00_ACLK M00_ARESETN M01_ACLK M01_ARESETN M02_ACLK M02_ARESETN M03_ACLK M03_ARESETN M04_ACLK M04_ARESETN

rgbleds_gpio

AXI GPIO

S_AXI s_axi_aclk GPIO
s_axi_aresetn

rgbleds_6bits

rpi_gpio

rst_ps7_0_fclk0

Processor System Reset

slowest_sync_clk ext_reset_in

aux_reset_in mb_debug_sys_rst dcm_locked

mb_reset bus_struct_reset[0:0] peripheral_reset[0:0]
interconnect_aresetn[0:0] peripheral_aresetn[0:0]

rst_ps7_0_fclk1

Processor System Reset

slowest_sync_clk ext_reset_in

aux_reset_in mb_debug_sys_rst dcm_locked

mb_reset bus_struct_reset[0:0] peripheral_reset[0:0]
interconnect_aresetn[0:0] peripheral_aresetn[0:0]

rst_ps7_0_fclk3

Processor System Reset

slowest_sync_clk ext_reset_in

aux_reset_in mb_debug_sys_rst dcm_locked

mb_reset bus_struct_reset[0:0] peripheral_reset[0:0]
interconnect_aresetn[0:0] peripheral_aresetn[0:0]

simple_io_0

simple_io_v1.0 (Pre-Production)

S00_AXI s00_axi_aclk GPIO[31:0] s00_axi_aresetn

switches_gpio

AXI GPIO

S_AXI GPIO
s_axi_aclk
s_axi_aresetn

ip2intc_irpt

sws_2bits

xlconcat_0

Concat

In0[0:0]
In1[0:0]

In2[0:0] dout[2:0]

axi_dynclk_0

Dynamic Clock Generator (Pre-Production)

s00_axi REF_CLK_I PXL_CLK_O

PXL_CLK_5X_O
LOCKED_O

s00_axi_aclk
s00_axi_aresetn

axi_smc

AXI SmartConnect

S00_AXI aclk M00_AXI aresetn

axi_vdma_0

AXI Video Direct Memory Access

S_AXI_LITE M_AXI_MM2S M_AXIS_MM2S
s_axi_lite_aclk m_axi_mm2s_aclk m_axis_mm2s_aclk axi_resetn mm2s_frame_ptr_in[5:0]

mm2s_frame_ptr_out[5:0] mm2s_introut

axis_subset_converter_0

AXI4-Stream Subset Converter

S_AXIS
aclk M_AXIS
aresetn

rgb2dvi_0

RGB to DVI Video Encoder (Source) (Pre-Production)

TMDS RGB
oen

aRst_n PixelClk SerialClk

v_axi4s_vid_out_0

AXI4-Stream to Video Out

video_in
vtiming_in

vid_io_out aclk aclken

aresetn
fid
vid_io_out_clk
vid_io_out_ce
vid_io_out_reset

vtg_ce
locked
overflow
underflow

fifo_read_level[12:0] status[31:0] sof_state_out

v_tc_0

Video Timing Controller

ctrl

vtiming_out

clk
clken

s_axi_aclk
s_axi_aclken
gen_clken
sof_state
resetn

s_axi_aresetn

irq

fsync_in

fsync_out[0:0]

xlconstant_0

Constant

dout[0:0]

DDR